Multivariable Calculus @ JHU 2026

Lecture: 3

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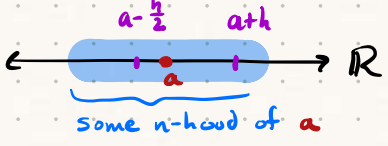
Topics: Differentials, Directional Derivatives, and Applications

Section I: Differentiability

Our first task is to cook up a viable definition of differentiability for functions $f:\mathbb{R}^n \to \mathbb{R}^m$ (the case where $f:\Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is handled in the same way, as we'll remark later). First, we recall—and then reconceive—the definition of differentiability often given for single variable functions.

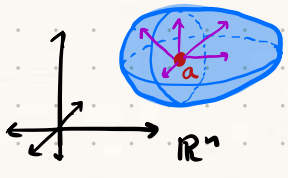
Definition 3.1 A function $f:\mathbb{R}\to\mathbb{R}$ is differentiable at the point $a\in\mathbb{R}$ iff $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists and is finite. In that case, the limit is unique and denoted by $f'(a)$.

In seeking to extend this notion to higher dimensions, we are met with a new phenomenon: in dim $n=1$, all of the points in a neighborhood of $a\in\mathbb{R}$ are of the form $a+h$ for $h\in\mathbb{R}$ small



so that the difference quotient $[f(a+h)-f(a)]/h$ considers all the different ways that $f(x)$ could vary in a n-hood of $a\in\mathbb{R}$ just by varying h.

In dim $n\geq 2$, a n-hood of a point $a\in\mathbb{R}^n$ has many more directions in which you can vary $f(x)$ when starting from a.



This alerts us that we need to be careful in our definition, and that we must consider all possible ways to approach a when defining the derivative for $f:\mathbb{R}^n\to\mathbb{R}^m$.

The adjustment, of course, is to let $h \in \mathbb{R}^n$ so that $a+h$ can explore the entire area of $\mathbb{R}^n$ around a.

An issue thus arises, however, if we try to directly generalize the single variable definition of differentiability above: if $f(a+h)-f(a) \in \mathbb{R}^m$ and $h \in \mathbb{R}^n$, then their quotient is not defined! Thus, our next adjustment could be to take a norm before taking the quotient:

$$\frac{f(a+h)-f(a)}{|h|} \qquad (*)$$

Considering the behavior of this quotient as $|h| \to 0$ thus captures something about the infinitesimal deviation of $f(x)$ from $f(a)$ near a, but in general this limit is not well behaved! For example, consider a friendly function with different rates of change in different directions, like $f: \mathbb{R}^2 \to \mathbb{R}$ given by $f(x,y) = x^2+y$. Let's consider $(*)$ at $(0,0)$:



i) if $h = te_1$ and $t \to 0$, $\dfrac{f(te_1,0)-f(0,0)}{|te_1|} = \dfrac{t^2}{|t|} \to 0$

whereas

ii) if $h = te_2$ and $t \to 0$, $\dfrac{f(0,te_2)-f(0,0)}{|te_2|} = \dfrac{t}{|t|} \to \pm 1$

Hence, $\lim\limits_{(h^1,h^2)\to\vec{0}} \dfrac{f(h^1,h^2)-f(0,0)}{|(h^1,h^2)|}$ does not exist, even though $f(x)$ seems perfectly nice at $(0,0)$! The error is evident though: because $f(x)$ can have different rates of change in different directions (as in i), ii) above), we can't expect a derivative of $f: \mathbb{R}^n \to \mathbb{R}^m$ to be a single vector. Instead, $f'(a)$ needs to account for the n-dimensional space of directions in which $f(x)$ can vary.

Side Remark: $(*)$ is actually really useful in the form $\lim\limits_{h\to 0} \dfrac{|f(a+h)-f(a)|}{|h|}$,

as it captures the largest possible rate of change of f at a.

Key Observation: What is a derivative really supposed to do? It captures the 1st order behavior of $f(x)$ near a: the best linear approximation to $f(x)$ near $a \in \mathbb{R}$ is given by $L_{f,a}(x) := f(a) + f'(a)\cdot(x-a)$. Writing $x = a+h$ and inserting this into the difference quotient, we see that indeed

$$\frac{f(a+h)-f(a)}{h} - f'(a) = \frac{f(a+h)-f(a)-f'(a)\cdot h}{h} = \frac{f(a+h)-L_{f,a}(a+h)}{h}$$

so that as $h \to 0$ and $x = a+h \to a$, $f(x)$ and $L_{f,a}(x)$ really do agree to 1st order at a when $f(x)$ is differentiable at a. At the heart of this approximation is the key idea that we lose something if we just think of $f'(a)$ as a number.

Instead, we should think of the derivative of $f(x)$ at a as the linear mapping $h \longmapsto f'(a)\cdot h$ for all $h \in \mathbb{R}$, which is the closest linear map approximating $f(x)-f(a)$ near a.

This formulation of the derivative is indeed sensitive to the different directions in which $f(x)$ can change, and it enables us to make the following (correct) generalization of the derivative to high dimensions:

Definition 3.2: A function $f: \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at the point $a \in \mathbb{R}$ iff there is a linear transformation $L: \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{|f(a+h)-f(a)-L(h)|}{|h|} = 0.$$

In this case, L is unique, and we write $L(-) = Df_a(-)$.

Note: This notion of differentiability is called Fréchet Differentiability.

Proof of uniqueness: If L_1 and L_2 both satisfy the limit condition above, then for any $x \in \mathbb{R}^n \setminus \{0\}$ and any $t \neq 0$,

$$\frac{|L_1(x) - L_2(x)|}{|x|} = \frac{|L_1(tx) - L_2(tx)|}{|tx|} \qquad \text{(linearity and } t \in \mathbb{R}\text{)}$$

$$\leq \frac{|L_1(tx) - (f(a+tx) - f(a))|}{|tx|} \qquad (\Delta\text{-ineq})$$

$$+ \frac{|(f(a+tx) - f(a)) - L_2(tx)|}{|tx|}$$

$$\longrightarrow 0 \quad \text{as } t \to 0. \qquad \text{(defn)}$$

Hence, $|L_1(x) - L_2(x)| = 0$ for every $x \in \mathbb{R}^n$ (including $x = 0$), so $L_1 = L_2$.

Let's revisit our previous example $f(x,y) = x^2 + y$ to show that it actually is differentiable at $(x,y) = (0,0)$ with $Df_{(0,0)} \approx [0,1]$ (here, we recall that the linear map $Df_{(0,0)} : \mathbb{R}^2 \to \mathbb{R}$ is represented by a (1×2) matrix acting on column vectors $h = (h^1; h^2)$). In a moment, we will give a method that makes it clear that $Df_{(0,0)}$ should be given by the matrix $[0,1]$, but for now let's just check the definition:

$$\frac{|f(h^1,h^2) - f(0,0) - Df_{(0,0)}(h^1,h^2)|}{|(h^1,h^2)|} = \frac{|(h^1)^2 + h^2 - (0,1)\cdot\begin{pmatrix} h^1 \\ h^2 \end{pmatrix}|}{\sqrt{(h^1)^2 + (h^2)^2}} = \frac{|(h^1)^2 + h^2 - h^2|}{\sqrt{(h^1)^2 + (h^2)^2}}$$

$$= \frac{(h^1)^2}{\sqrt{(h^1)^2 + (h^2)^2}} \leq \frac{(h^1)^2}{|h^1|} \leq |h^1| \xrightarrow{h \to \vec{0}} 0.$$

Notice also: $Df_{(0,0)} \approx [0,1]$ encodes the rates of change of f in the different coordinate directions as in (i), (ii) above, and it's transpose also happens to point in the direction of in which $f(x,y)$ increases the fastest at $(0,0)$.

Oftentimes, the differential $Df_a(-)$ is also called the Jacobian of f at a, although some authors distinguish between them as

- the differential $Df_a(-)$ as the abstract linear map
- the Jacobian $f'(a)$ as the $(m \times n)$ matrix representing $Df_a(-)$ with respect to the standard bases on $\mathbb{R}^n, \mathbb{R}^m$. If different bases are chosen, then $f'(a)$ changes accordingly, while $Df_a(-)$ is the same!

n.b. in this course we will typically use the standard bases s.t. $Df_a(v) = f'(a) \cdot v$

Note: In 1-D, the number $f'(a)$ is the entry in the (1×1)-matrix $[f'(a)]$ acting on $\mathbb{R}^1$, representing the linear map $h \longmapsto f'(a)h$.

Remark on Domains:

In the above disussions, we considered functions $f: \mathbb{R}^n \to \mathbb{R}^m$. This was (and will be) just a notational convenience. More often, one considers functions $f: \Omega \to \mathbb{R}^m$ for some $\Omega \subseteq \mathbb{R}^n$.

In case Ω is open, differentiability of f at $a \in \Omega$ is the same as above, since openness implies that $a + h \in \Omega = \text{Domain}(f)$ for all small enough $h \in \mathbb{R}^n$. We say that $f: \Omega \to \mathbb{R}^m$ is differentiable on Ω if it is differentiable at every $a \in \Omega$.

In case Ω is not open, and $a \in \Omega$ is not an interior point, differentiability at a means that f can be extended to some larger open set $\widetilde{\Omega} \supseteq \Omega$ (that is there is an extension function $\widetilde{f}: \widetilde{\Omega} \to \mathbb{R}^m$ with $\widetilde{f} = f$ on Ω) with a in the interior of $\widetilde{\Omega}$, and with $\widetilde{f}$ differentiable in the usual sense at a. In this situation, f is differentiable on Ω if $\widetilde{f}$ is differentiable on $\widetilde{\Omega}$.

Example: (This sort of case shows up very often in PDE settings, for example) Let $f: \overline{B_1(0)} \to \mathbb{R}$ be given by $f(x) = |x|^2$. f is differentiable on $\overline{B_1(0)}$, since (for example) f extends to $B_2(0) \supseteq \overline{B_1(0)}$ by the same formula, and $\widetilde{f}(x) = |x|^2$ on $B_2(0)$ is indeed differentiable at all $x \in \overline{B_1(0)}$.

Fact: This procedure is well-defined, in the sense that the actual differential at a non-interior point obtained in this way does not depend on the choice of extension $\widetilde{f}: \widetilde{\Omega} \to \mathbb{R}^m$. (Check this!)

Section II: Derivative Rules

In this section, we will formulate and prove the basic rules of differential calculus in our new high dimensional setting. To avoid distracting technicalities about domains of definition, we'll assume throughout that our functions are globally defined.

We begin with the

Theorem 3.3: (Chain Rule)

Let $f: \mathbb{R}^n \to \mathbb{R}^m$ be diff. at $a \in \mathbb{R}^n$, and $g: \mathbb{R}^m \to \mathbb{R}^k$ diff. at $f(a)$. Then $g \circ f: \mathbb{R}^n \to \mathbb{R}^k$ is diff. at a with

$$D(g \circ f)_a = Dg_{f(a)} \circ Df_a$$

Remark: On the level of Jacobians (i.e. matrices representing these linear maps), the result says that $(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$ ← matrix multiplication.

Proof: To ease the notation, lets set $b = f(a)$, $T = Df_a$, $S = Dg_{f(a)}$, and write

$$\varphi(x) = f(x) - f(a) - T(x-a)$$
$$\xi(y) = g(y) - g(b) - S(y-b)$$
$$p(x) = (g \circ f)(x) - (g \circ f)(a) - S(T(x-a))$$

By assumption,

$$\lim_{x \to a} \frac{|\varphi(x)|}{|x-a|} = \lim_{y \to b} \frac{|\xi(y)|}{|y-b|} = 0$$

and we must show that

$$\lim_{x \to a} \frac{|p(x)|}{|x-a|} = 0.$$

To this end, we first do some algebra:

$$p(x) = g(f(x)) - g(b) - S(T(x-a)) = g(f(x)) - g(b) - S(f(x) - f(a) - \varphi(x))$$

$$= g(f(x)) - g(b) - S(f(x) - b) + S(\varphi(x))$$
$$= \xi(f(x)) + S(\varphi(x))$$

Thus, it will suffice to prove that

$$\lim_{x \to a} \frac{|\xi(f(x))|}{|x-a|} = \lim_{x \to a} \frac{|S(\varphi(x))|}{|x-a|} = 0 .$$

① ②

First of all, recall that since T and S are linear maps on a f.d. vector space, they are bounded. Hence, there are constants C_T and C_S such that

$$|T(v)| \leq C_T |v| \quad \forall v \in \mathbb{R}^n \qquad \text{and} \qquad |S(w)| \leq C_S |w| \quad \forall w \in \mathbb{R}^m .$$

① Since $\lim_{y \to b} \frac{|\xi(y)|}{|y-b|} = 0$, if we pick any $\varepsilon > 0$ there will exist a $\delta = \delta(\varepsilon) > 0$ s.t. $|y-b| < \delta \Rightarrow |\xi(y)| < \varepsilon |y-b|$. In particular, as long as $|f(x) - b| < \delta$, $|\xi(f(x))| < \varepsilon |f(x) - b|$. Since $f(x)$ is continuous at $x = a$*, $\exists \eta = \eta(\delta) > 0$ s.t. $|x-a| < \eta \Rightarrow |f(x) - b| < \delta$. Therefore, as long as $|x-a| < \eta$, we have

$$|\xi(f(x))| < \varepsilon |f(x) - b|$$

which gives...

$$= \varepsilon |T(x-a) + \varphi(x)|$$
$$\leq \varepsilon \cdot |T(x-a)| + \varepsilon \cdot |\varphi(x)| .$$
$$\leq \varepsilon \cdot C_T |x-a| + \varepsilon |\varphi(x)|$$

Hence,

$$\lim_{x \to a} \frac{|\xi(f(x))|}{|x-a|} \leq \varepsilon \cdot C_S ,$$

and since $\varepsilon > 0$ was arbitrary, ① follows.

② This one is easier:

$$\frac{|S(\varphi(x))|}{|x-a|} \leq \frac{C_S |\varphi(x)|}{|x-a|} \to 0 \quad \text{as } x \to a .$$

* We will prove this in Theorem 3.8 below!

Next, we prove several important facts about differentials, and establish an auxilliary result that will help prove the sum and product rules.

Theorem 3.4:

(i) If $f:\mathbb{R}^n \to \mathbb{R}^m$ is constant (i.e. $f(x) = y_0$ for all $x \in \mathbb{R}^n$), then $Df_a = 0$ for all $a \in \mathbb{R}^n$ (i.e. the linear map $v \mapsto 0 \cdot v = 0$)

(ii) If $f:\mathbb{R}^n \to \mathbb{R}^m$ is linear, then $Df_a = f$ for all $a \in \mathbb{R}^n$.

(iii) If $f:\mathbb{R}^n \to \mathbb{R}^m$ has $f(x) = (f^1(x), f^2(x), \ldots, f^m(x))$, then f is diff. at $a \in \mathbb{R}^n$ iff each component $f^i:\mathbb{R}^n \to \mathbb{R}$ is, and moreover

$$Df_a = \left((Df^1)_a, \ldots, (Df^m)_a\right)$$

– i.e. the matrix of Df_a is the $(m \times n)$-matrix

$$f'(a) = \begin{bmatrix} - (f^1)'(a) - \\ \vdots \\ - (f^m)'(a) - \end{bmatrix} \qquad \left(i^{th}\ \text{row} = \underset{(f^i)'(a)}{(1\times n)\text{-matrix}}\right)$$

(iv) (Tool to prove sum rule)

Define $s:\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ by $s(x,y) = x + y$. Then s is diff. and

$$(Ds)_{(a,b)} = S \qquad \forall (a,b) \in \mathbb{R}^n \times \mathbb{R}^n$$

(v) (Tool to prove product rule)

Define $p:\mathbb{R}^2 \to \mathbb{R}$ by $p(x,y) = x \cdot y$. Then p is diff and

$$Dp_{(a,b)}(x,y) = bx + ay \qquad \forall (a,b) \in \mathbb{R}^2.$$

Proof: (i) $|f(a+h) - f(a) - 0 \cdot h| \equiv |y_0 - y_0| \equiv 0$

(ii) $|f(a+h) - f(a) - f(h)| \equiv |f(a) + f(h) - f(a) - f(h)| \equiv 0$

(indeed a linear transformation is its own best linear approximation!)

(iii) ($\Rightarrow$) Let $\pi^i:\mathbb{R}^m \to \mathbb{R}$ be the projection onto the ith coordinate

$$\pi^i(x) = \pi^i(x^1, \ldots, x^m) = x^i \qquad \text{(Check this is linear!)}$$

Then $f^i(x) = \pi^i \circ f(x)$, and $(D\pi^i)_a = \pi^i$ at every $a \in \mathbb{R}^m$ by (ii).

Thus by the Chain Rule, $f^i(x)$ is differentiable if $f(x)$ is, and

$$D(f^i)_a = \pi^i \circ (Df)_a$$

$D(f)_a : \mathbb{R}^n \to \mathbb{R}^m$ is represented by an $(m \times n)$ matrix, and this equation reads that the ith row of that matrix is the $(1 \times n)$ matrix (row vector) representing the differential $D(f^i)_a : \mathbb{R}^n \to \mathbb{R}$.

($\Leftarrow$) Suppose each $f^i = \pi^i \circ f$ is diff. at $a \in \mathbb{R}^n$, and define the linear transformation.

$$T := \left(D(f^1)_a, \dots, D(f^m)_a\right) : \mathbb{R}^n \to \mathbb{R}^m$$

Then

$$\frac{|f(a+h) - f(a) - T(h)|}{\|h\|} = \frac{|(\dots, f^i(a+h) - f^i(a) - D(f^i)_a(h), \dots)|}{|h|}$$

$$\leq \sum_{i=1}^m \frac{\|f^i(a+h) - f^i(a) - D(f^i)_a(h)\|}{|h|} \qquad (\Delta\text{-ineq})$$

Sending $h \to 0$ proves that f is diff. at a with $Df_a = T$. ▨

(iv) $S : \mathbb{R}^2 \to \mathbb{R}$ is linear, so this follows by (ii).

(v) Let $T(x,y) = bx + ay$ and compute

$$\frac{|p(a+h^1, b+h^2) - p(a,b) - T(h^1,h^2)|}{\|h\|} = \frac{|ab + bh^1 + ah^2 + h^1h^2 - ab - bh^1 - ah^2|}{\sqrt{(h^1)^2 + (h^2)^2}}$$

$$= \frac{|h^1h^2|}{\sqrt{(h^1)^2 + (h^2)^2}} \leq \frac{(h^1)^2 + (h^2)^2}{\sqrt{(h^1)^2 + (h^2)^2}} = \sqrt{(h^1)^2 + (h^2)^2}$$

which tends to 0 as $h \to 0$. ▨

Remark: By induction, one can extend (iv) and (v) to arbitrary finite sums and products, of course.

Using (iv) and (v) above, we can prove the sum and product rules:

Corollary 3.5: Suppose $f:\mathbb{R}^n \to \mathbb{R}^m$ and $g:\mathbb{R}^n \to \mathbb{R}^m$ are both diff. at $a \in \mathbb{R}^n$. Then

(i) $f+g$ is diff. at a with $D(f+g)_a = Df_a + Dg_a$.

(ii) If $g:\mathbb{R}^n \to \mathbb{R}^1$, then $g \cdot f$ is diff. at a with

$$D(gf)_a(-) = (Dg_a)(-)\cdot f(a) + g(a)\cdot Df_a(-)$$
$$= (\ldots, f^i(a)\cdot Dg_a + g(a)\cdot D(f^i)_a, \ldots)$$

(iii) If $g:\mathbb{R}^n \to \mathbb{R}^1$ has $g(a) \neq 0$, then f/g is diff. at a with

$$D(f/g)_a(-) = \frac{g(a)\cdot Df_a(-) - Dg_a(-)\cdot f(a)}{g(a)^2}.$$

Proof: (i) This follows from the Chain Rule and (iv), since $f+g = S(f,g)$:

$$D(f+g)_a = DS_{(f(a)+g(a))} \circ (D(f,g)_a)$$
$$= S \circ (Df_a, Dg_a) = Df_a + Dg_a$$

(ii) This follows from the Chain Rule, (iii), (iv), since $g\cdot f = g\cdot(f^1,\ldots,f^m)$ $= (p(g,f^1),\ldots,p(g,f^m))$:

$$D(gf)_a = (D[p(g,f^1)]_a, \ldots, D[p(g,f^m)]_a)$$
$$= (Dp_{(g(a),f^1(a))} \circ (Dg_a, D(f^1)_a), \ldots)$$
$$= ((f^1(a), g(a)) \cdot (Dg_a, D(f^1)_a), \ldots)$$
$$= (f^1(a)\cdot Dg_a(-) + g(a)\cdot D(f^1)_a(-), \ldots).$$

(iii) This follows from (ii), the Chain Rule, and $D[\frac{1}{x}]_{a\neq 0} = -\frac{1}{a^2}$.

Exercises/Examples: Compute the differentials of:

1. $f(x,y) = \sin(xy)$, $f(x,y,z) = (x, e^{yz})$, $f(x) = (x^2, x^3)$
2. A bilinear function $T:\mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^k$.
3. $f^{-1}:\mathbb{R}^m \to \mathbb{R}^n$, for a diff. $f:\mathbb{R}^n \to \mathbb{R}^m$ with diff. inverse.

Section III: Directional Derivatives

Let's return to our first discussion on possible extensions of differentiability to high dimensions. So far, we have been setting down the theory of Fréchet Differentiability for functions $f: \mathbb{R}^n \to \mathbb{R}^m$, which captures the sense in which $f(x)$ has approximately linear behavior near a point. In this section, we will introduce another notion of differentiability called Gateaux Differentiability. In general, this is a weaker notion of differentiability than Fréchet Diff, in that while Fréchet Diff. implies Gateaux Diff, the converse can fail (a Gateaux Diff function might even be discontinuous!). Nevertheless, the Gateaux derivative is extremely useful, is used ubiquitously, and can imply Fréchet Diff. in certain contexts, as we will see.

Definition 3.6: Let $f: \mathbb{R}^n \to \mathbb{R}^m$, and $a \in \mathbb{R}^n$ a basepoint. Fix also a direction vector $v \in \mathbb{R}^n$. Then f is Gateaux Diff. at a in the direction v iff the following limit exists and is finite:

$$\lim_{t \to 0} \frac{f(a+tv) - f(a)}{t} = \frac{d}{dt}\Big|_{t=0} f(a+tv) \in \mathbb{R}^m.$$

In this case, the limit is called a Gateaux (or directional) derivative of f and is a function of a and v denoted by $D_v f_a$.

If $D_v f_a$ exists for every $v \in \mathbb{R}^n$, we say that f is Gateaux Differentiable at a, and call $D_{(-)} f_a : \mathbb{R}^n \to \mathbb{R}^m$ the Gateaux differential of f at a. Exercise: Check this is well-defined!

Remark:

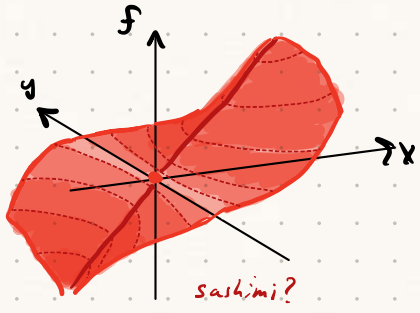
Since the directional derivative is the more commonly used term for this concept in vector calculus, that's what we will use most of the time. Technically, the terms "Fréchet derivative" and "Gateaux derivative" come

from the generalization of these objects to even more general situations including the infinite dimensional setting.... but that's a matter for a functional analysis course! At any rate---

Gateaux Diff. at $a \in \mathbb{R}^n$ implies that there exists a map $D_{G)} f_a : \mathbb{R}^n \to \mathbb{R}^m$ with the property that

$$\lim_{t \to 0} \frac{|f(a+tv) - f(t) - t\, D_v f a|}{t} = 0 \quad \text{for all } v \in \mathbb{R}^n,$$

but NOTE that this does not imply that f is Fréchet Diff at a, and nor is $v \longmapsto D_v f_a$ a linear map in general! Here is an example illustrating how $D_v f_a$ is calculated, and how it's behavior might be nonlinear:



Example: Define $f(x,y) = \begin{cases} \frac{x^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$.

Fix a direction vector $v = (a,b) \neq 0$. Then for $t \neq 0$,

$$\frac{f(tv) - f(0,0)}{t} = \frac{1}{t} \cdot \frac{t^3 a^3}{t^2 a^2 + t^2 b^2} = \frac{a^3}{a^2+b^2}$$

so that $D_v f_{(0,0)} = \frac{a^3}{a^2+b^2}$. Note: we identify the column vector $\begin{pmatrix} \frac{a^3}{a^2+b^2} \end{pmatrix}$ in $\mathbb{R}^1$ with the scalar $\frac{a^3}{a^2+b^2}$ here, as is customary when computing the directional derivatives of scalar valued functions, which is usually what is meant when writing $f: \mathbb{R}^n \to \mathbb{R}$... In any case, the map $v = (a,b) \longmapsto D_v f_{(0,0)} = \frac{a^3}{a^2+b^2}$ is defined for all $v \in \mathbb{R}^n$, so that f is Gateaux Diff. at (0,0). However, $D_{G)} f_{(0,0)} : \mathbb{R}^n \to \mathbb{R}$ is not linear in v! As we will see in a moment, this prevents f from being Fréchet Diff. at (0,0).

There is a very important "special case" of the above construction, when $f: \mathbb{R}^n \to \mathbb{R}$ is scalar valued and we take $v \in \{e_1, \ldots, e_n\}$. Conventionally, we usually identify $\mathbb{R}^1$ (1×1 vectors) with $\mathbb{R}$ (scalars) here:

Definition 3.7: Let $f:\mathbb{R}^n \to \mathbb{R}$ and fix $a \in \mathbb{R}^n$. The limit

$$\lim_{h \to 0} \frac{f(a^1, \ldots, a^i + h, \ldots, a^n) - f(a^1, \ldots, a^n)}{h},$$

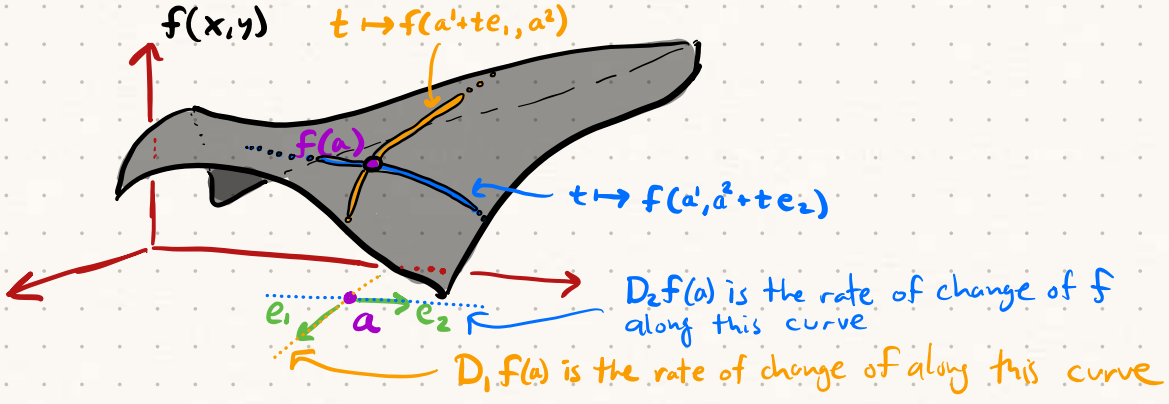
if it exists, is called the ith partial derivative of $f(x)$ at a, and is denoted by any of the following notations:

$$\frac{\partial f}{\partial x^i}(a) = \partial_{x^i} f(a) = \partial_i f(a) = D_i f(a) = D_{e_i} f_a$$

This is just the normal derivative of the 1-variable function $f(a^1, \ldots, \overset{i^{th}\text{ place}}{x}, \ldots, a^n)$ (x = i^{th} coordinate) at the point $x = a^i$. Hence, if $h(x) = (a^1, \ldots, x, \ldots, a^n)$, $D_i f(a) = (f \circ h)'(a^i)$.

We want to emphasize the suggestiveness of the notation $D_i f(a)$, since it will often be the case that we think of the partials as scalar valued functions $x \longmapsto D_i f(x)$ themselves.

Geometrically, we have the following sort of picture:



Again, for a vector valued function $f:\mathbb{R}^n \to \mathbb{R}^m$, $D_v f_a \in \mathbb{R}^m$ is a vector describing the best linear approximation to f in the direction v thru the point a. For scalar valued function $f:\mathbb{R}^n \to \mathbb{R}$ (such as the component functions of a vector valued map), the partial derivative $D_i f(a)$ is the scalar rate of instantaneous change of f in the ith coordinate direction at a.

Examples/Exercises: Compute all the partial derivatives of the following:

① $f(x,y) = \sin(xy)$ ② $f(x,y,z) = (x, e^{yz})$ ③ $f(x) = (x^2, x^3)$

and compare with the Jacobians computed for these functions previously.

Section IV: Differentiability Conditions

In this section we'll address the following question:

Q: How does the existence of partial derivatives / Gateaux Diff. at a point relate to Fréchet Diff.?

Let $f: \mathbb{R}^n \to \mathbb{R}^m$ and fix $a \in \mathbb{R}^n$.

First observation: Even if all the partials $D_i f_a$ exist, f need not be Gateaux Diff. at a.

Example: Let $f(x,y) = \begin{cases} 1 & (x,y) \in \{x\text{-axis}\} \cup \{y\text{-axis}\} \\ 0 & \text{else} \end{cases}$

Then $D_1 f_{(0,0)} = D_2 f_{(0,0)} = 0$, but no other $D_v f_{(0,0)}$ exists!

Second Observation: Even if f is Gateaux Diff. at a, f need not be continuous at a, let alone Fréchet Diff.

Example: Let $f(x,y) = \begin{cases} 0 & \{y=0\} \\ x^3/y & \{y \neq 0\} \end{cases}$

Then f is Gateaux Diff at $(0,0)$ since $\forall v = (v^1, v^2) \in \mathbb{R}^2$

$$\frac{f(tv) - f(0,0)}{t} = \frac{f(tv)}{t} = \begin{cases} 0 & \{v^2 = 0\} \\ (tv^1)^3 / t(tv^2) = t\frac{(v^1)^3}{v^2} & \{v^2 \neq 0\} \end{cases}$$

so that $D_v f_{(0,0)} = 0$ for every $v \in \mathbb{R}^2$.

Nevertheless, f is not even continuous at $(0,0)$, since eg) $f(t, t^3) \to 1$ as $t \to 0$, while $f(0,0) = 0$.

Thus, f cannot be Fréchet Diff at $(0,0)$ either, since

Theorem 3.8: If $f: \mathbb{R}^n \to \mathbb{R}^m$ is Fréchet Diff. at a, then f is continuous at a.

Proof: Let $0<|h|<\varepsilon$. Then

$$|f(a+h)-f(a)| \leq |h|\frac{|f(a+h)-f(a)-Df_a(h)|}{|h|} + |Df_a(h)|$$

and for ε small enough we can thus make $|f(a+h)-f(a)| \leq \delta$ for any chosen $\delta>0$. Thus, $f(x)\to f(a)$ when $x\to a$. ▨

Together, these examples illustrate the following (subtle) fact:

It's possible for a function to "look linear" along every direction thru a point, while **not** looking linear overall at that point.

In particular, this is due to the fact that the **scale** at which f looks linear at a point in a certain direction might **depend upon that direction**: For a given $v\in\mathbb{R}^n$, t might need to be <u>much smaller</u> for $|f(a+tv)-f(a)|$ to be as small as it is for another direction $w\in\mathbb{R}^n$. Thus, there may not even exist a scale (size of h's) for which $f(a+h)-f(a)\approx 0$, let alone $f(a+h)-f(a)\approx$ linear in h.

Frēchet Diff. is thus much more restrictive on $f(x)$: it says that given any $\delta>0$, there exists a scale $\varepsilon>0$ so that for <u>any</u> $x\in\mathbb{R}^n$ so that $x=a+h$ with $|h|<\varepsilon$, we have $f(a+h)-f(a)\approx Df_a(h)$.

This discussions suggests that Frēchet Diff. should control directional derivatives. Indeed:

Theorem 3.9: Frēchet Diff. at $a \Rightarrow$ Gateaux Diff at a. Moreover, for all $v\in\mathbb{R}^n$, $D_v f_a = Df_a(v)$.

Proof: In the definition of (Frēchet) differentiability, just set $h=tv$ (take $v\neq 0$) and send $t\to 0$ to get

$$0=\lim_{t\to 0}\frac{|f(a+tv)-f(a)-Df_a(tv)|}{|tv|} = \lim_{t\to 0}\frac{1}{|v|}\left|\frac{f(a+tv)-f(a)}{t}-Df_a(v)\right|$$

hence it follows that $D_v f_a = Df_a(v)$. ▨

An important corollary of this is the following, which gives us an efficient way to find the Jacobian $f'(a)$ of Df_a if we know that f is differentiable at a:

Corollary 3.10: If $f:\mathbb{R}^n \to \mathbb{R}^m$ is Frechet Diff. at a, then all partial derivatives $D_i f^j(a)$ exist, and moreover $f'(a) = \left[D_i f^j(a)\right]_{\substack{i=1,\dots,n \\ j=1,\dots,m}}$ in the standard bases of $\mathbb{R}^n$, $\mathbb{R}^m$.

Proof: Existence follows from the preceding theorem, and the Jacobian entries can be verified by first checking that

$$(D_{e_i} f)_a = Df_a(e_i) = f'(a)\cdot e_i = i^{th} \text{ column of } f'(a).$$

where the penultimate equality is because of our decision to use the standard bases. Since $f = (f^1, f^2, \dots, f^m)$, can thus identify the jth entry of the column $(D_i f)_a$ as exactly $D_i f^j(a)$, since

$$(D_{e_i} f)_a = \lim_{t\to 0} \frac{f(a+te_i) - f(a)}{t} = \lim_{t\to 0} \left(\frac{f^1(a+te_i)-f^1(a)}{t}, \dots, \frac{f^m(a+te_i)-f^m(a)}{t} \right)$$
$$= \left(D_i f^1(a), \dots, D_i f^m(a) \right).$$

Alternatively, write $D_i f^j(a) = (\pi^j \circ f \circ h)'(a^i)$ where $h(x) = (a^1, \dots, \overset{i^{th}\text{ place}}{x}, \dots, a^n)$, and use the Chain Rule in Jacobian form:

$$\left[D_i f^j(a)\right] = (\pi^j)'(f(a)) \cdot f'(a) \cdot h'(a^i) = \overset{j^{th}\text{ spot}}{[0,\dots,1,\dots,0]} \cdot f'(a) \cdot \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}^{i^{th}\text{ spot}}. \qquad \blacksquare$$

Key Construction: When $f:\mathbb{R}^n \to \mathbb{R}$ is scalar valued and differentiable at a, we often use the notation

$$\nabla f(a) = \left(D_1 f(a), \dots, D_n f(a)\right) \in \mathbb{R}^n,$$

called the gradient of f. As a matrix, the column vector $\nabla f(a)$ is the transpose of the row vector $f'(a)$. By Theorem 3.9 above, for any vector $v \in \mathbb{R}^n$,

$$D_v f(a) = Df_a(v) = \langle \nabla f(a), v \rangle = \|\nabla f(a)\| \cdot \|v\| \cdot \cos\Theta(\nabla f(a), v)$$

Therefore, $\nabla f(a)$ is a vector pointing in the direction of fastest increase for f!

Examples/Exercises: Assuming that the following functions are Frechet Diff., find $f'(a)$:

① $f(x) = |x|^2$, $x \in \mathbb{R}^n$ ② $f(x,y) = (x^2, yx, e^y)$ ③ $f(x,y,z) = xyz$.

The above results and examples illustrate:



Naturally, the difficulty in calculating these objects decreases as we move left to right in the above diagram. It would thus be nice to know if there was a situation in which the existence of all partial derivatives implies something about the existence of Df_a. The next Theorem is such a result:

Theorem 3.11: Let $f: \mathbb{R}^n \to \mathbb{R}^m$ and fix $a \in \mathbb{R}^n$. If every $D_i f^j(x)$ exists for x in an open neighborhood of a, and if the functions $D_i f^j(x)$ are continuous at a, then f is Frechet Diff. at a (and of course, $f'(a) = [D_i f^j(a)]_{i=1,\dots,n}^{j=1,\dots,m}$).

Proof: The main tool in this proof is the Mean Value Theorem from single variable calculus:

Theorem (MVT): Suppose $\phi: [a,b] \to \mathbb{R}$ is continuous, and differentiable on (a,b). Then there exists a point $c \in (a,b)$ such that

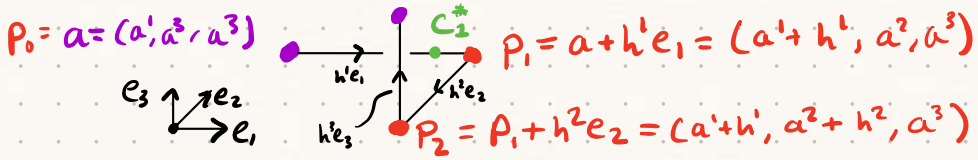
$$\phi(b) - \phi(a) = \phi'(c) \cdot (b-a).$$

For the sake of readability, we will make the simplifying assumption that $f: \mathbb{R}^n \to \mathbb{R}$ is scalar valued. Then, we can invoke the previous Theorem ensuring that $f: \mathbb{R}^n \to \mathbb{R}^m$ is diff. iff. its components are.

Proof of Theorem 3.11: Let's focus our attention in an open cube $Q_\varepsilon(a)$ on which all the $Df(x)$ exist. Fix $h = (h^1, \dots, h^n) \neq 0$ with $|h| < \varepsilon$ so that $a+h \in Q$ and consider all expressions

$$f(a+h) - f(a) = f(a^1+h^1, \dots, a^n+h^n) - f(a^1, \dots, a^n)$$

for such h. We will approximate this difference by writing this it as a sum of n "rectilinear jumps." To wit, consider the following path from a to $a+h$ (drawn in 3D to illustrate):



In general, we define the stepping stones

$$\begin{aligned} p_0 &= a \\ p_1 &= a + h^1 e_1 \\ &\vdots \\ p_k &= a + h^1 e_1 + \cdots + h^k e_k \equiv p_{k-1} + h^k e_k \\ &\vdots \\ p_n &= a + h^1 e_1 + \cdots + h^n e_n = a+h \end{aligned}$$

and notice that the path from p_k to p_{k+1} is a straight line segment parametrized by $\gamma_k(t) = p_{k-1} + t e_k = (a^1+h^1, \dots, a^{k-1}+h^{k-1}, a^k+t, a^{k+1}, \dots, a^n)$ for t going from 0 to h^k (note h^k could be 0, $+$, or $-$), which only varies the kth coordinate. Now write

$$\begin{aligned} f(a+h) - f(a) &= f(p_n) - f(p_0) \\ &= [f(p_n) - f(p_{n-1})] + [f(p_{n-1}) + f(p_{n-2})] + \cdots \\ &\qquad + [f(p_1) - f(p_0)] \end{aligned}$$

and notice that in each term we can write

$$f(p_k) - f(p_{k-1}) = f \circ \gamma_k(h^k) - f \circ \gamma_k(0)$$

so that, by the MVT, there is some t^k between 0 and h^k, corresponding to some point $c_k^* = \gamma_k(t^k)$ in the line segment from p_{k-1} to p_k (and thus $c_k^* \in Q_\varepsilon(a)$ with $|a - c_k^*| \leq |h|$) so that

$$f(p_k) - f(p_{k-1}) = f(\gamma_k(h^k 1)) - f(\gamma_k(0))$$
$$= (f \circ \gamma_k)'(c_k^*) \cdot (h^k - 0)$$
$$= (D_k f)(c_k^*) \cdot h^k$$

hence

$$f(a+h) - f(a) = \sum_{k=1}^{n} f(p_k) - f(p_{k-1})$$
$$= \sum_{k=1}^{n} (D_k f)(c_k^*) \cdot h^k .$$

Now we can finish the argument: for such $0 < |h| < \varepsilon$, $c_k^* \in Q_\varepsilon(a)$ as above, we estimate the difference quotient:

$$\frac{|f(a+h) - f(a) - f'(a) \cdot h|}{|h|} = \frac{|\sum_{k=1}^{n} (D_k f)(c_k^*) h^k - \sum_{k=1}^{n} (D_k f)(a) \cdot h^k|}{|h|}$$
$$\leq \sum_{k=1}^{n} |(D_k f)(c_k^*) - (D_k f)(a)| \cdot \frac{|h^k|}{|h|}$$
$$\leq \sum_{k=1}^{n} |(D_k f)(c_k^*) - (D_k f)(a)|$$

as $h \to 0$, all of the $c_k^* \to a$, and so by continuity of the $D_k f$ at a, we see that

$$\frac{|f(a+h) - f(a) - f'(a) \cdot h|}{|h|} \to 0 .$$



Important Consequence: This theorem tells us that we can rest assured of the differentiability of many commonly encountered functions by just checking their partial derivatives for existence and continuity, which is often much quicker than analyzing Frechet difference quotients!

Exercises / Examples: Show that the following functions are Frechet diff., and find their Jacobians, by computing all of the partial derivatives and checking for continuity:

1. $f(x) = \sin x$, $x_0 = 0$
2. $f(x,y,z) = (x, x+y, z)$, $x_0 = (1,1,1)$
3. $f(x,y) = \sin(xy)$, $x_0 = (\pi, \pi)$
4. $f(x,y,z,w) = (x^2 y^2, w \sin(z))$, $x_0 = (1,1,\frac{\pi}{2},0)$